

CFTC Positioning Pattern Thresholds — Experiment Results for Sign-Off

1. Purpose

Grid-search results for **SQUEEZE** (FM low + RM high) and **LIQUIDITY EXIT** (RM low + FM high) before locking production Sentiment Layer 3 flags. **Display/alert only** — not SSI sizing gates (per 2 Aug email you sent).

2. Data & methodology

Item	Value
CFTC Fast Money through	2026-07-28 (Tuesday position date)
CFTC Real Money through	2026-07-28
Percentile window	156 weeks (~3 years), rolling
Backtest start	2006-01-01
Forward returns	S&P 500 at 4w / 8w / 12w trading days
Grid step	5 percentile points
SQUEEZE JSON	03_squeeze_grid_20260804.json
LIQUIDITY EXIT JSON	04_liquidity_exit_grid_20260804.json

SQUEEZE: FM pctile < X AND RM pctile > Y (same week).

LIQUIDITY EXIT: RM pctile < X AND FM pctile > Y.

LIQUIDITY EXIT **4w down %** = share of episodes with negative 4w SPX return. SQUEEZE **12w win %** = share with positive 12w SPX return.

3. Executive summary

- SQUEEZE** is bullish context — best Sharpe (n≥50): **FM<20, RM>45** (n=125, 12w avg +3.3174%, Sharpe 1.1752, win 77.5%).
- PDF default FM<30/RM>50: n=170, 12w avg +2.6634%, Sharpe 0.8782 — valid but less sharp than FM<20/RM>45.
- LIQUIDITY EXIT** RM<30/FM>60: n=116, 4w SPX-down 35.34%, 12w avg +2.8429% — modest stress flag, not a strong short.
- Patterns are **common** (~5-10 fires/year) — use as context flags only.

Recommended options

Pattern	Option A (sharper)	Option B (PDF / more frequent)
SQUEEZE	FM < 20 , RM > 45 (Sharpe 1.1752)	FM < 30 , RM > 50 (n=170, Sharpe 0.8782)

Pattern	Option A (sharper)	Option B (PDF / more frequent)
LIQUIDITY EXIT	RM < 25 , FM > 55 (see §5)	RM < 30 , FM > 60 (n=116, 4w down 35.34%)

4. SQUEEZE heatmap — 12w avg SPX % / Sharpe

FM <	RM>40	RM>45	RM>50	RM>55	RM>60	RM>65
15	3.0529% / 1.057	3.1777% / 1.0925	3.1825% / 1.07	3.0791% / 1.0215	3.1236% / 1.0097	3.034% / 0.9398
20	3.2279% / 1.1526	3.3174% / 1.1752	3.3295% / 1.1619	3.2581% / 1.1212	3.2367% / 1.0806	3.1561% / 1.0182
25	3.0045% / 1.0184	3.0704% / 1.0324	3.0781% / 1.0139	3.0121% / 0.9808	2.8851% / 0.903	2.7436% / 0.8333
30	2.6982% / 0.9123	2.7143% / 0.9105	2.6634% / 0.8782	2.5981% / 0.8491	2.3708% / 0.7478	2.1971% / 0.6698
35	2.5421% / 0.7988	2.5515% / 0.7954	2.5287% / 0.7748	2.4262% / 0.7356	2.2238% / 0.6511	2.0415% / 0.5763

Top SQUEEZE cells by 12w Sharpe (n ≥ 50)

FM <	RM >	n	4w avg	4w win %	12w avg	12w win %	12w Sharpe
20	45	125	0.5872	63.2	3.3174	77.5	1.1752
20	50	118	0.7415	65.25	3.3295	77.88	1.1619
20	40	130	0.6246	64.62	3.2279	76.0	1.1526
20	55	114	0.7815	65.79	3.2581	77.06	1.1212
15	45	92	0.4542	61.96	3.1777	78.16	1.0925
20	60	107	0.7665	65.42	3.2367	75.49	1.0806
15	50	85	0.6575	64.71	3.1825	78.75	1.07
15	40	96	0.5151	63.54	3.0529	75.82	1.057
25	45	152	0.5561	65.79	3.0704	77.4	1.0324
15	55	82	0.716	65.85	3.0791	77.92	1.0215
25	40	157	0.5881	66.88	3.0045	76.16	1.0184
20	65	99	0.8143	65.66	3.1561	73.4	1.0182

5. LIQUIDITY EXIT — top cells (RM ≤ 35, FM ≥ 45)

RM <	FM >	n	4w avg	4w down %	12w avg	Median DD
35	45	172	0.8228	33.14	2.7615	-0.0161
35	50	160	0.6588	35.0	2.4619	-0.4434

RM <	FM >	n	4w avg	4w down %	12w avg	Median DD
30	45	156	0.7126	33.33	2.9085	0.0996
35	55	146	0.6401	34.25	2.5948	-0.0811
30	50	145	0.5789	35.17	2.6529	-0.0563
25	45	134	0.7417	34.33	3.1469	0.1485
30	55	131	0.5496	34.35	2.8214	0.0525
35	60	129	0.611	35.66	2.6794	-0.0563
25	50	124	0.5837	36.29	2.8128	-0.0811
30	60	116	0.5472	35.34	2.8429	0.0492

PDF default RM<30, FM>60

- **4w:** n=116, avg 0.5472%, down 35.34%, Sharpe 0.407
- **8w:** n=116, avg 1.3558%, down 37.07%, Sharpe 0.5679
- **12w:** n=116, avg 2.8429%, down 29.31%, Sharpe 0.8341

6. Sign-off

SQUEEZE: FM < __ , RM > __ (A: 20/45 · B: 30/50)

LIQUIDITY EXIT: RM < __ , FM > __ (A: 25/55 · B: 30/60)

8. Spot check (2026-08-04)

Latest COT position date **2026-07-28**: FM pctile **67.3**, RM pctile **60.9**.

Pattern	Threshold	Fires today?
SQUEEZE	FM<20, RM>45	No
SQUEEZE	FM<30, RM>50	No
LIQUIDITY EXIT	RM<25, FM>55	No
LIQUIDITY EXIT	RM<30, FM>60	No

Neither pattern active at current prints — consistent with mid-range positioning.

9. Reproduce

```
cd MindWealth_UI
.venv/bin/python -c "from src.sentiment_superindex.analysis.cftc_grid import run_and_report;
run_and_report('2006-01-01')"
.venv/bin/python scripts/compile_cftc_pattern_threshold_report.py
```